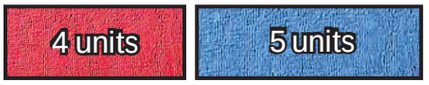
Situation 3:

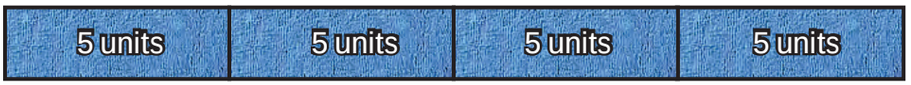
Consider the red and the blue coloured floor mats of length 4 units and 5 units as follows.



Five red coloured floor mats of 4 units each can be arranged as follows. Its total length is 5 x 4 = 20 units.



Four blue coloured floor mats of 5 units each can be arranged as follows. Its total length is also the same 4 x 5 = 20 units.



Note that the 5 floor mats each of length 4 units are required to equal 4 floor mats each of length 5 units and that is, the length 20 units is the smallest common length that can be matched by both sizes. From the above, it shows that the least common multiple of 4 and 5 is 4x5=20.

The **Least Common Multiple** of any two non-zero whole numbers is the smallest or the lowest common multiple of both the numbers. The Least Common Multiple of the numbers *x* and *y* can be written as LCM (*x,y*).

We can find the least common multiple of two or more numbers by the following methods.

1. **Division Method**
2. **Prime Factorisation Method**

Example 5: Find the LCM of 156 and 124.

Solution: By **Division method**

Step 1: Start with the smallest prime factor and go on dividing till all the numbers are divided as given below.

2	156,124
2	78, 62
3	39, 31
13	13, 31
31	1, 31
	1,1

Step 2: LCM = product of all prime factors

= 2 x 2 x 3 x 13 x 31 = 4836

Thus, the LCM of 156 and 124 is 4836.

By **Prime Factorisation method**

Step 1: We write the prime factors of 156 and 124 as given below (use of divisibility test rules will also help).

156 = 2 x 78 = 2 x 2 x 39 = 2 x 2 x 3 x 13

124 = 2 x 62 = 2 x 2 x 31

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